Entropic Open-set Active Learning

AAAI, 2024

이미지 처리팀

강인하 안가영 **안종식** 장보아 최승준 황장훈

CONTENTS

01 | Introduction

02 | Related Works

03 | Methods

04 | Experiments

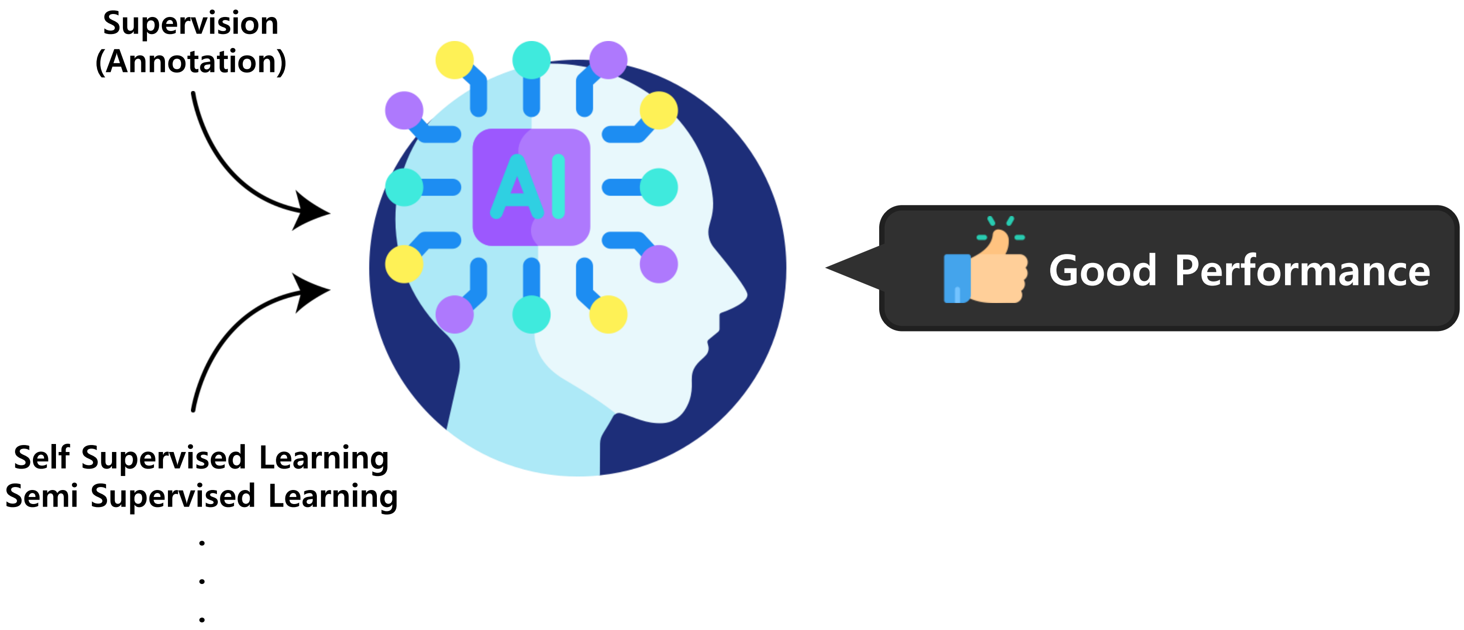
05 | Conclusion

01

Introduction

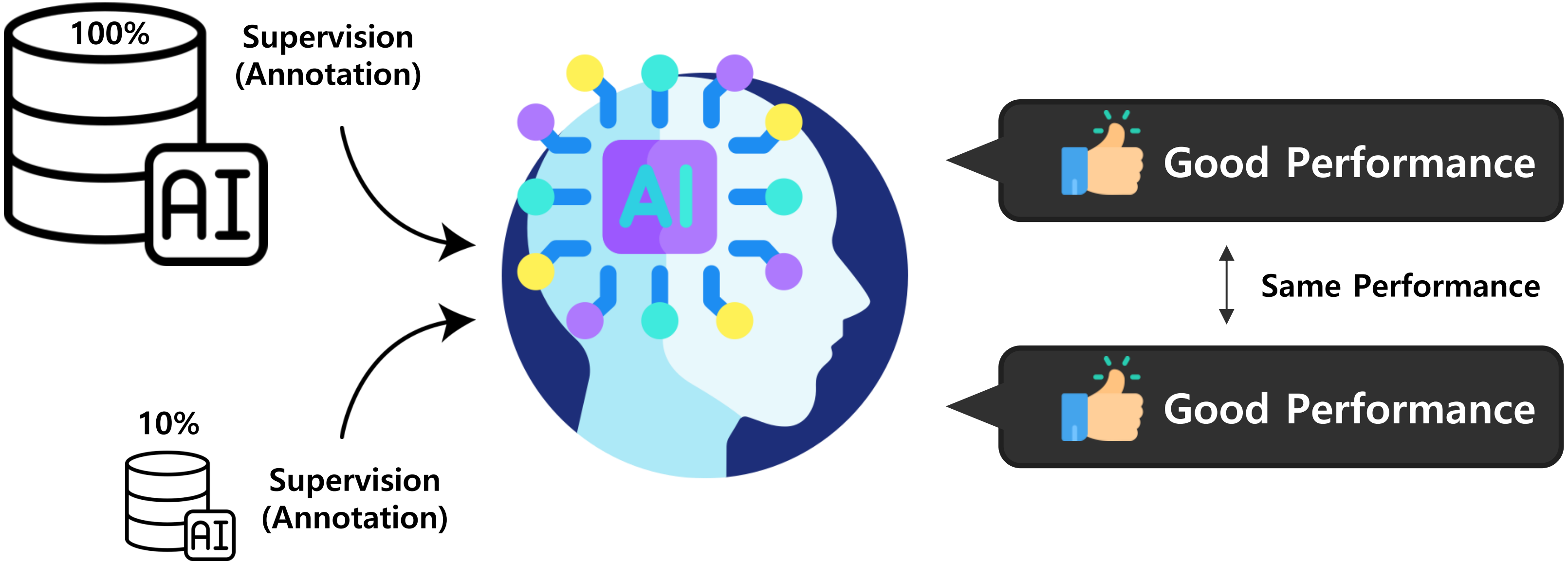
01. Introduction

1. Active Learning (AL)



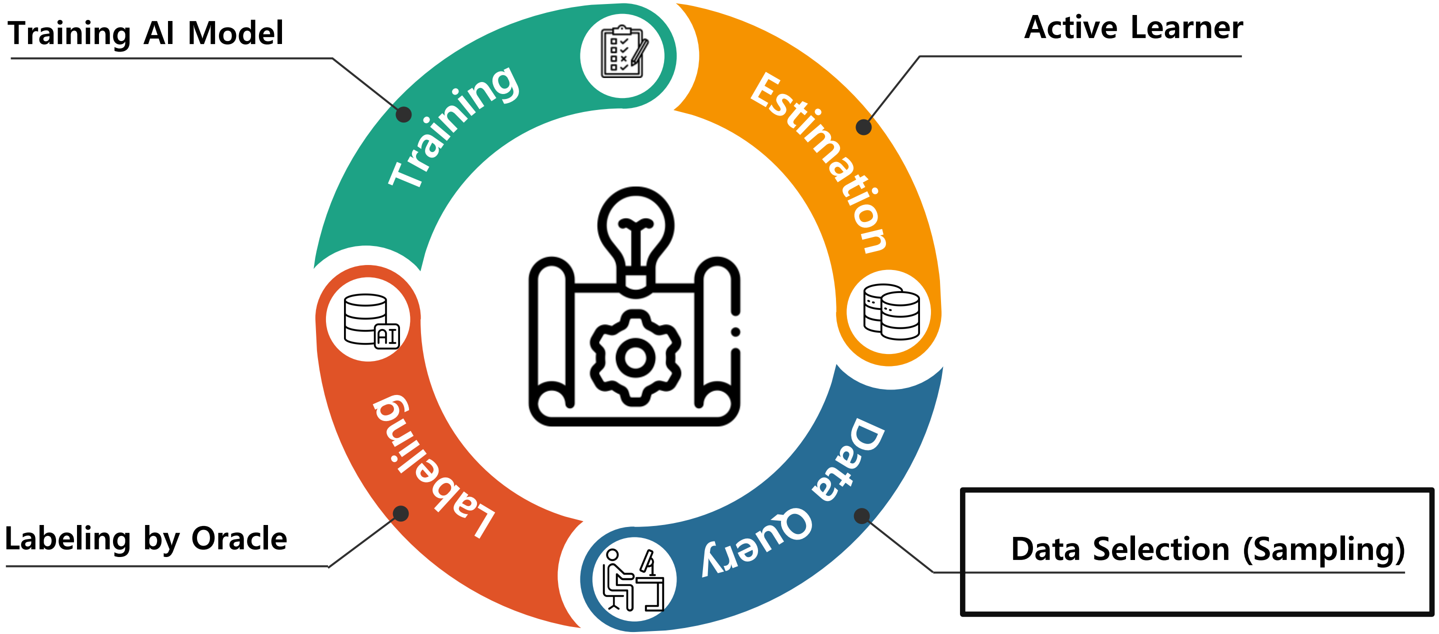
01. Introduction

1. Active Learning (AL)



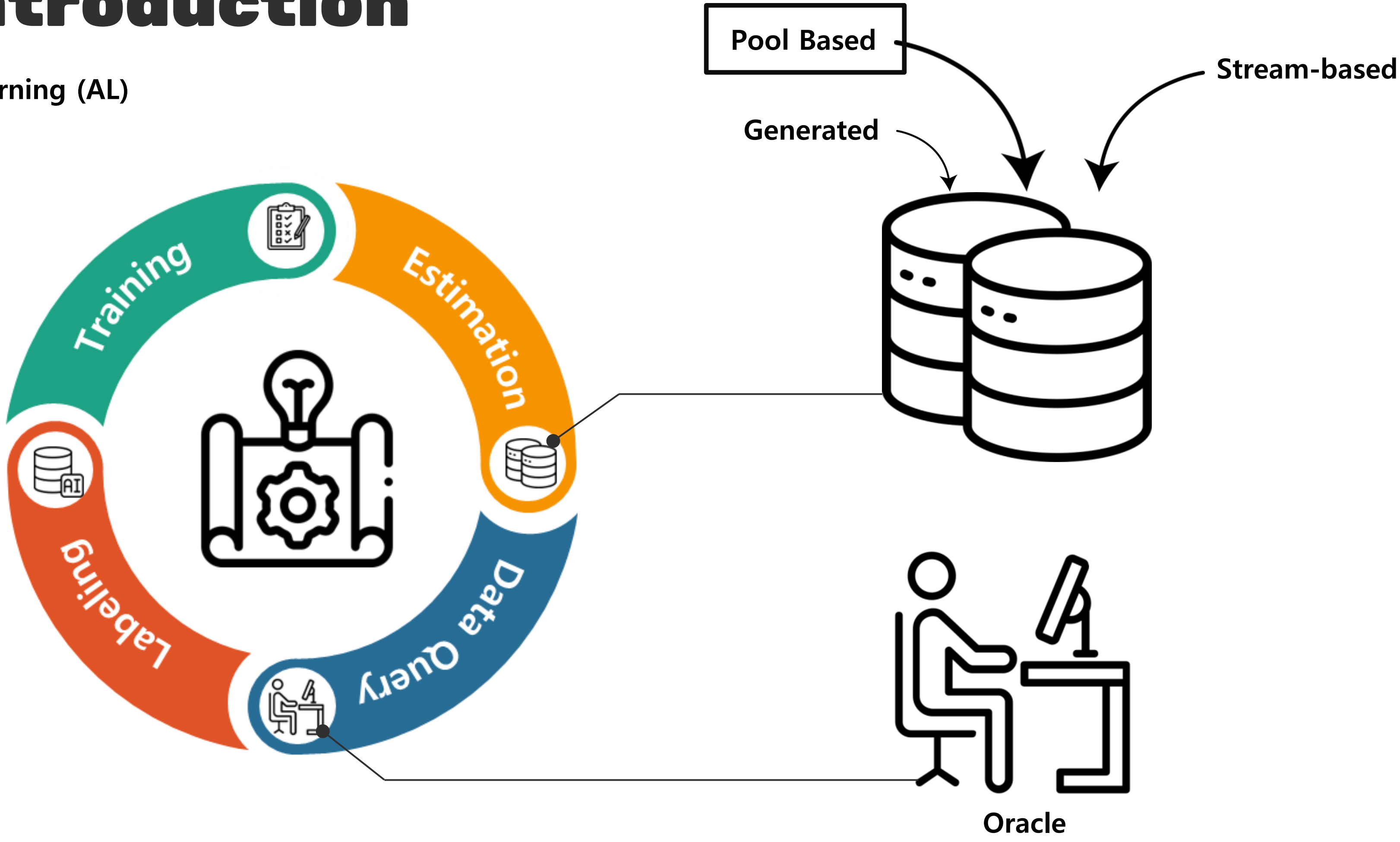
01. Introduction

1. Active Learning (AL)



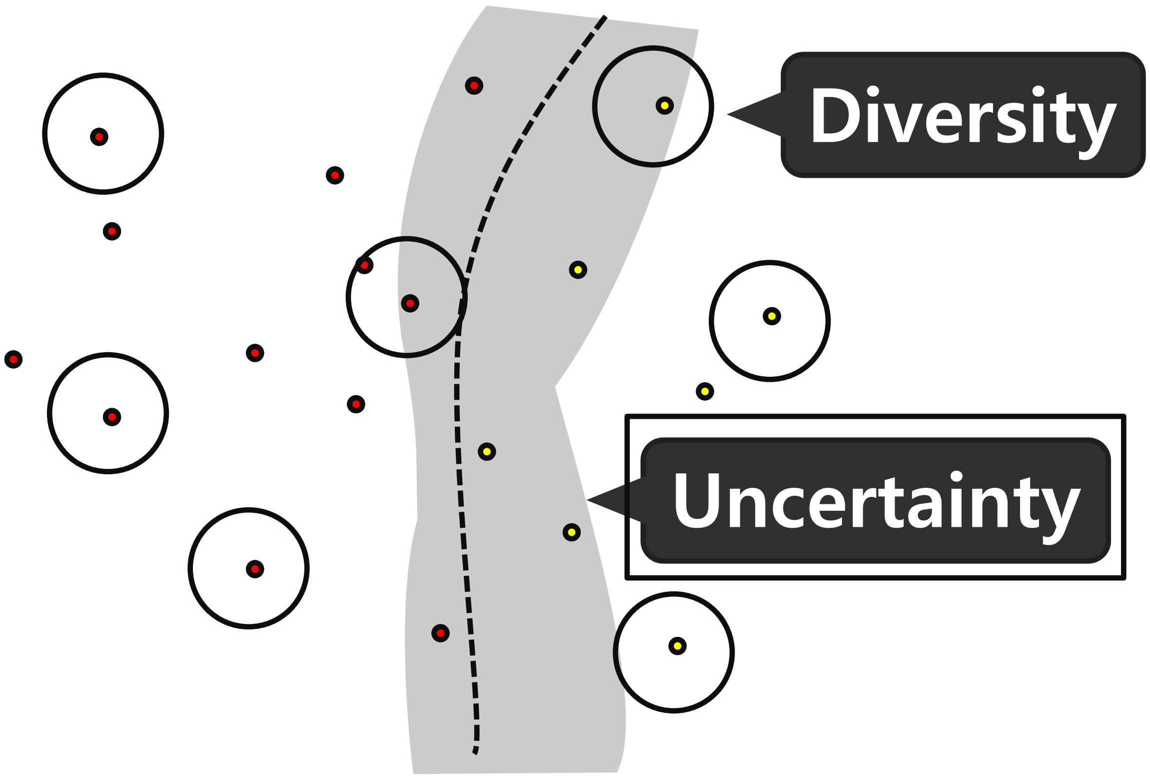
01. Introduction

1. Active Learning (AL)



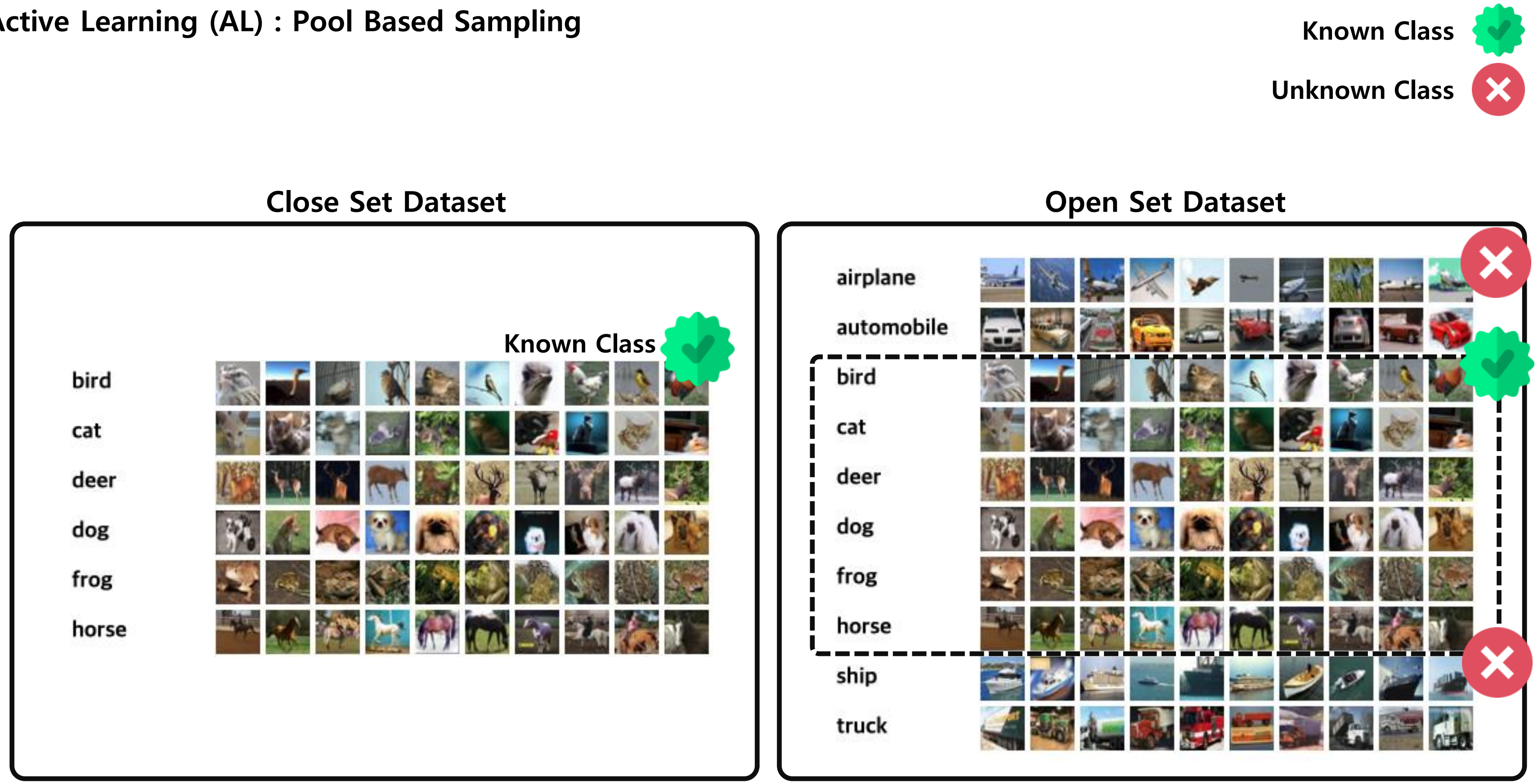
01. Introduction

1. Active Learning (AL) : Pool Based Sampling



01. Introduction

1. Active Learning (AL) : Pool Based Sampling



02

Related Works

02. Related Works

1. Active Learning

- **Main Result Comparison Method**

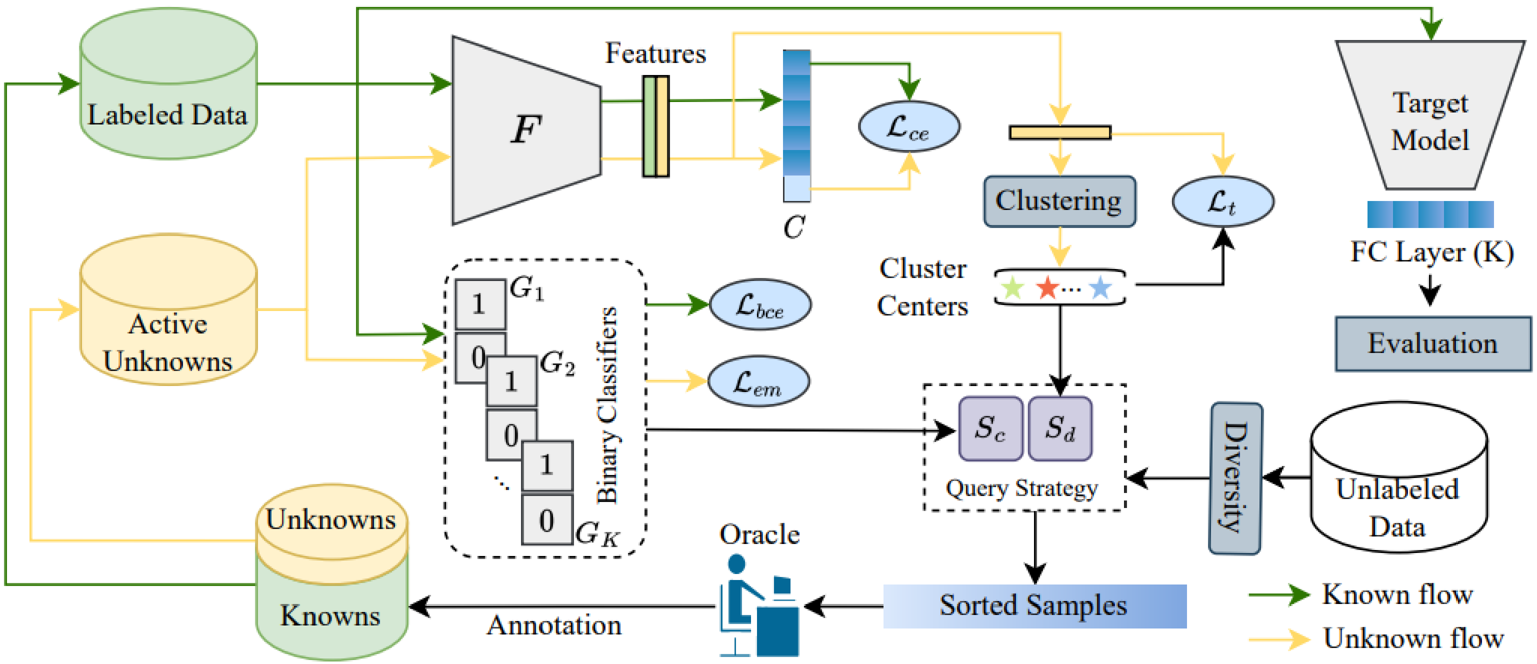
구분	방식	Method	설명
Standard AL	Diversity	Random Sampling	무작위 Samling 방법
	Uncertainty	Entropy	모델의 예측 Entropy를 사용하여 샘플 선택
		Certainty	모델의 Confidence를 기반으로 선별
		BALD (Bayesian Active Learning by Disagreement)	샘플들이 서로 다른 예측 시 정보량이 높다는 개념으로 샘플 선택
	Diversity	Coreset	데이터 대표성을 가지는 샘플들을 선택하여 학습 효율성 향상
Open Set AL	Uncertainty	MQNet	Open set AL에서 Purity-informativeness의 Trade-off를 해결하여 중복 데이터 제거 등을 통해 많은 데이터를 수집하는데 초점을 둔 논문
		LfOSA	Known Class의 Purity를 최대화하여 Known Sample과 Unknown Sample을 구별하는 논문

03

Methods

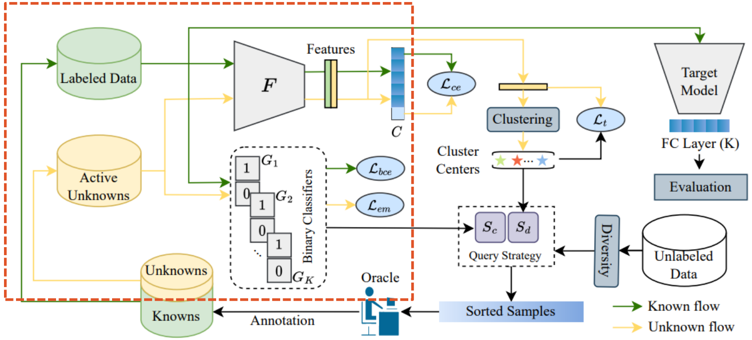
03. Methods

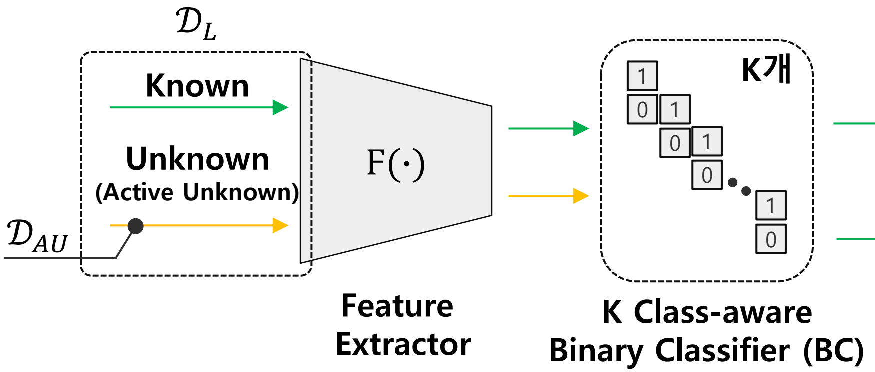
1. EOAL[Entropic Open-set Active Learning] Overview



03. Methods

2. Close-set Entropy Scoring





Binary Cross Entropy Loss

$$\mathcal{L}_{bce} = \frac{1}{n_l} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_L} -\log(p^{y_i}) - \min_{j \neq y_i} \log(1 - p^j)$$

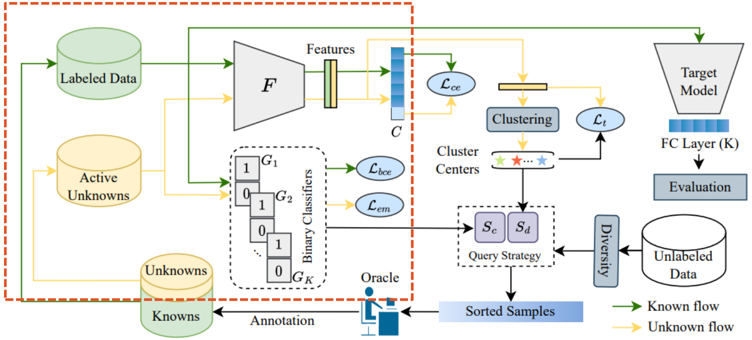
$$H_i(\mathbf{x}) = -p^i \cdot \log(p^i) - (1 - p^i) \cdot \log(1 - p^i)$$

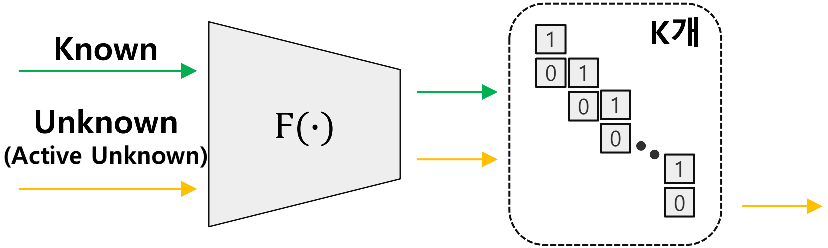
Close-set Entropy Score

$$S_c(\mathbf{x}) = \frac{1}{K \cdot \log(2)} \sum_{i=1}^{K} H_i(\mathbf{x})$$

03. Methods

2. Close-set Entropy Scoring





Feature Extractor

K Class-aware Binary Classifier (BC)

Close-set Entropy Score

$$S_c(\mathbf{x}) = \frac{1}{K \cdot \log(2)} \sum_{i=1}^{K} H_i(\mathbf{x})$$

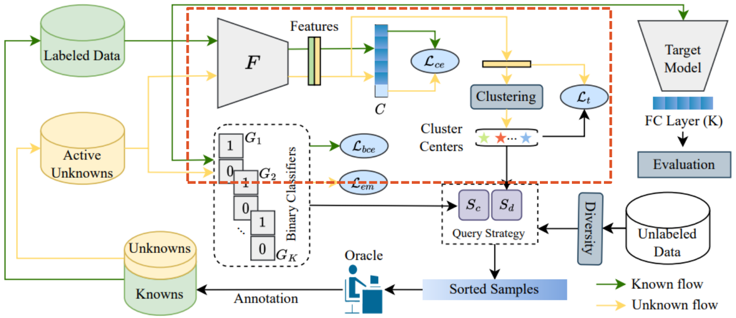
$$\mathcal{L}_{em} = \frac{1}{K \cdot n_{au}} \sum_{\mathbf{x} \in \mathcal{D}_{AU}} \sum_{i=1}^{K} -\frac{1}{2}\log(p^i) - \frac{1}{2}\log(1-p^i)$$

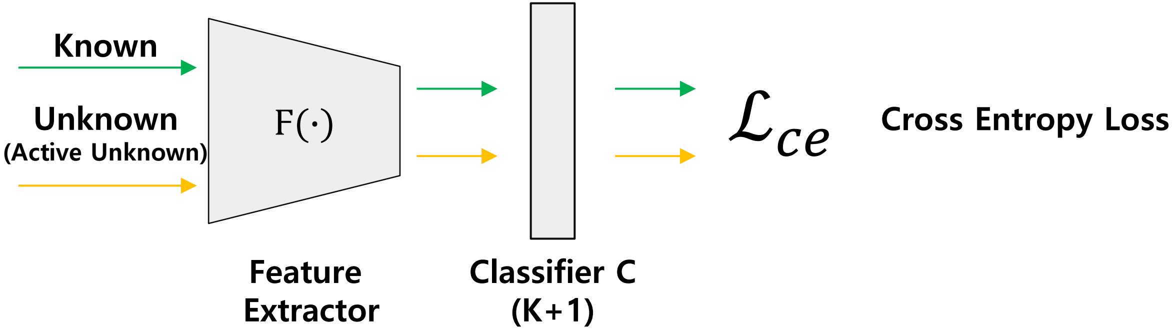
Entropy Maximize(?) Loss

Property 1. *Minimizing L_{em} is equivalent to maximizing the entropy of each G_i.*
Proof. We have $p^i + (1 - p^i) = 1$, and $p^i \in (0, 1)$. By applying Jensen's inequality for concave functions, we obtain $H_i(\mathbf{x}) \leq \log(2)$ and $\mathcal{L}_{em} \geq \log(2)$, where the equality happens *iff* $p^i = (1 - p^i) = \frac{1}{2}$.

03. Methods

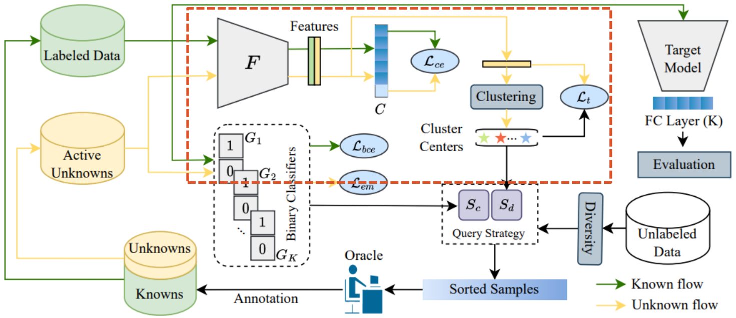
3. Distance based Entropy Scoring

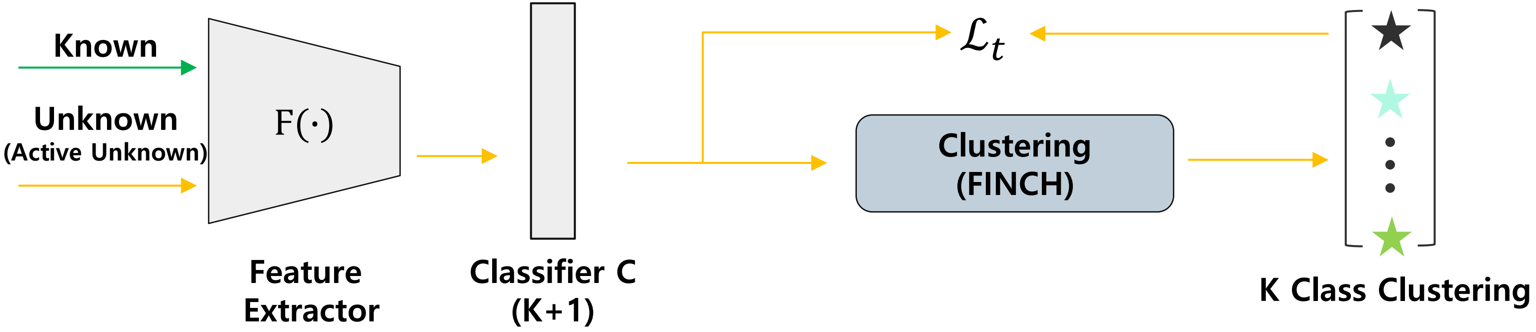




03. Methods

3. Distance based Entropy Scoring



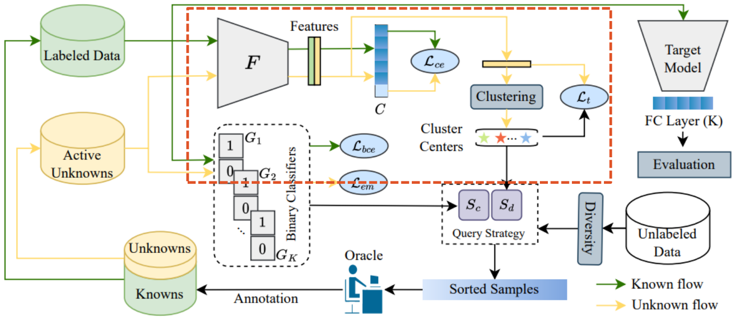


Tuplet loss

$$\mathcal{L}_t = \frac{1}{n_{au}} \sum_{\mathbf{x} \in \mathcal{D}_{AU}} \log\left(1 + \sum_{j \neq \hat{y}}^{K} e^{D_{\hat{y}} - D_j}\right) + \beta D_{\hat{y}} \quad \text{where } D_i = \|F(\mathbf{x}) - \mathbf{c}_i\|$$

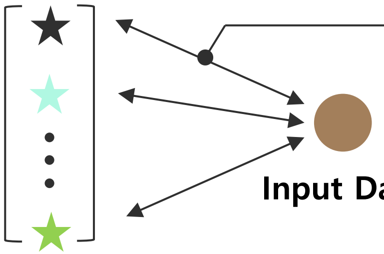
03. Methods

3. Distance based Entropy Scoring



Belonging Probability of i-th Cluster

$$q_i(\mathbf{x}) = \frac{e^{-\|F(\mathbf{x})-\mathbf{c}_i\|/T}}{\sum_{j=1}^{K} e^{-\|F(\mathbf{x})-\mathbf{c}_j\|/T}}$$



$$S_d(\mathbf{x}) = \frac{-1}{\log(K)} \sum_{i=1}^{K} q_i(\mathbf{x}) \cdot \log(q_i(\mathbf{x}))$$

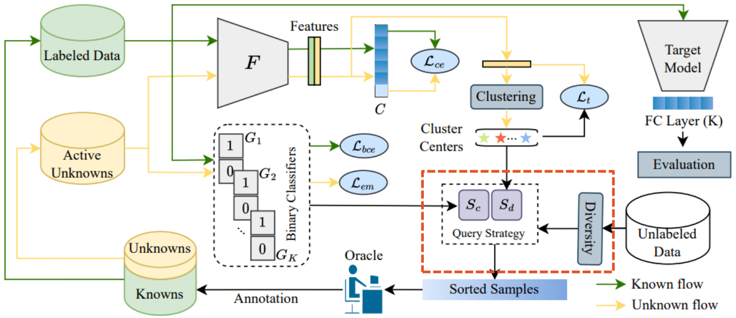
Distance Based Entropy

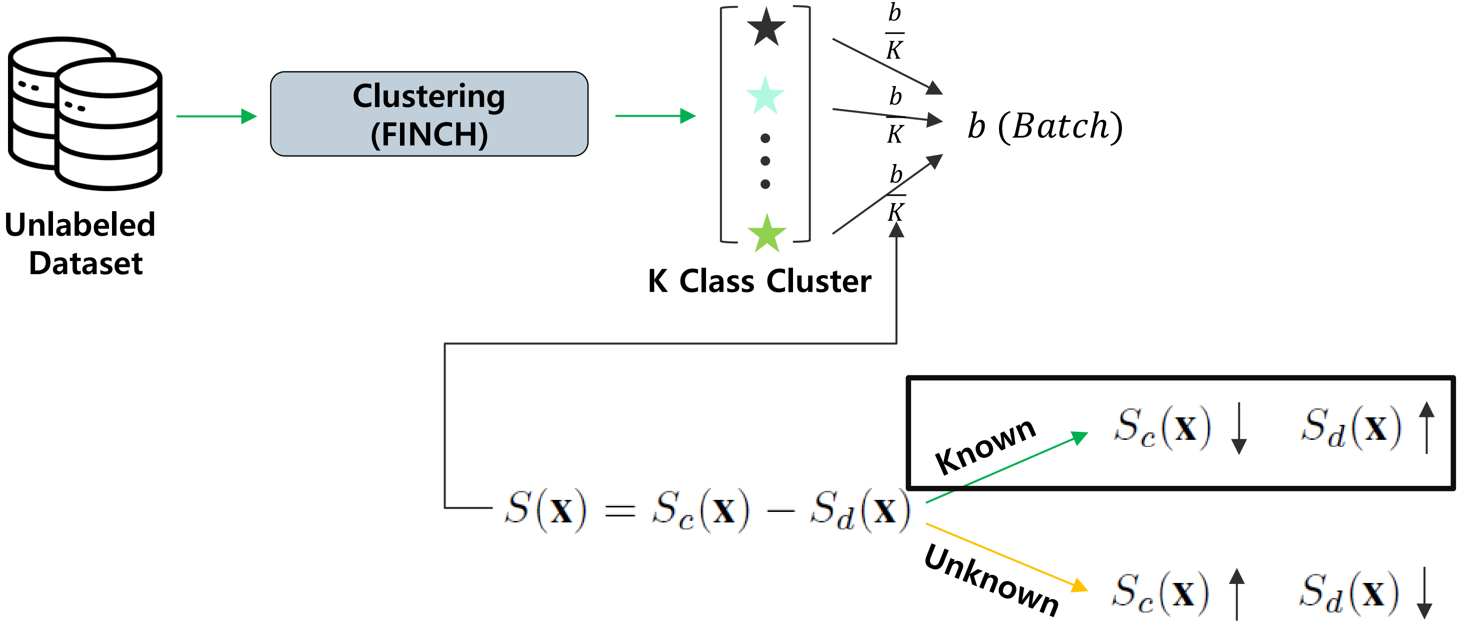
K Class Cluster Centers

$$\mathbf{c}_i = \frac{\sum_{(\mathbf{x},\hat{y})\in\mathcal{D}_{AU}} \mathbb{I}\{\hat{y}=i\} \cdot F(\mathbf{x})}{\sum_{(\mathbf{x},\hat{y})\in\mathcal{D}_{AU}} \mathbb{I}\{\hat{y}=i\}}$$

03. Methods

4. Query Strategy





03. Methods

5. Flow

Algorithm 1: Our Proposed Algorithm for Open-set AL

Input:
  Labeled data $\mathcal{D}_L$, unlabeled data $\mathcal{D}_U$, number of AL cycles R, known categories K, per-cycle budget b, models F, C, T, and $\{G_i\}_{i=1}^{K}$
Process:
  $\mathcal{D}_{AU} \leftarrow \emptyset$ *# Initial active unknowns*
  Update models F, C, and $\{G_i\}_{i=1}^{K}$ by minimizing $\mathcal{L}_{total}$ in Eq. 9

최초 소량의 Labeling Data로 학습

  for $c = 0, 1, ..., R-1$ **do**
    $\forall x \in D_U, S_d(x) \leftarrow 0$ *# Initialization*
    if $\mathcal{D}_{AU} \neq \emptyset$ **do**
      Cluster the features of $\mathcal{D}_{AU}$ into K clusters
      For cluster i, compute the center $\mathbf{c}_i$ using Eq. 5
      $\forall x \in D_U$, compute $S_d(x)$ via Eq. 6
    end if
    $\forall x \in D_U$, compute $S_c(x)$ via Eq. 2
    Cluster the features of $\mathcal{D}_U$ into K clusters, $\{C_1, C_2, ..., C_K\}$ *# Diversity*
    for $j = 1, 2, ..., K$ **do**
      $\mathcal{S}_j \leftarrow \{S_c(x) - S_d(x) | \forall x \in C_j\}$ *# Uncertainty*
      $X_j \leftarrow$ select the $\frac{b}{K}$ samples with the lowest values from $\mathcal{S}_j$, and annotate them
    end
    # All queries for the current cycle:
    $X^{active} \leftarrow X_1 \cup X_2 \cup ... \cup X_K$
    # Knowns and active unknowns:
    Obtain X^k and X^u
    $\mathcal{D}_L \leftarrow \mathcal{D}_L \cup X^k$, $\mathcal{D}_{AU} \leftarrow \mathcal{D}_{AU} \cup X^u$,
    $\mathcal{D}_U \leftarrow \mathcal{D}_U / X^{active}$ *# Update datasets*
    Update the target model T via minimizing the cross-entropy loss on D_L
  end

03. Methods

5. Flow

Algorithm 1: Our Proposed Algorithm for Open-set AL

Input:
  Labeled data $\mathcal{D}_L$, unlabeled data $\mathcal{D}_U$, number of AL cycles R, known categories K, per-cycle budget b, models F, C, T, and $\{G_i\}_{i=1}^{K}$
Process:
  $\mathcal{D}_{AU} \leftarrow \emptyset$ *# Initial active unknowns*
  Update models F, C, and $\{G_i\}_{i=1}^{K}$ by minimizing $\mathcal{L}_{total}$ in Eq. 9
  for $c = 0, 1, ..., R-1$ **do**
    $\forall x \in D_U, S_d(x) \leftarrow 0$ *# Initialization*
    if $\mathcal{D}_{AU} \neq \emptyset$ **do**
      Cluster the features of $\mathcal{D}_{AU}$ into K clusters
      For cluster i, compute the center $\mathbf{c}_i$ using Eq. 5
      $\forall x \in D_U$, compute $S_d(x)$ via Eq. 6
    end if
    $\forall x \in D_U$, compute $S_c(x)$ via Eq. 2
    Cluster the features of $\mathcal{D}_U$ into K clusters, $\{C_1, C_2, ..., C_K\}$ *# Diversity*
    for $j = 1, 2, ..., K$ **do**
      $\mathcal{S}_j \leftarrow \{S_c(x) - S_d(x) | \forall x \in C_j\}$ *# Uncertainty*
      $X_j \leftarrow$ select the $\frac{b}{K}$ samples with the lowest values from $\mathcal{S}_j$, and annotate them
    end
    # All queries for the current cycle:
    $X^{active} \leftarrow X_1 \cup X_2 \cup ... \cup X_K$
    # Knowns and active unknowns:
    Obtain X^k and X^u
    $\mathcal{D}_L \leftarrow \mathcal{D}_L \cup X^k$, $\mathcal{D}_{AU} \leftarrow \mathcal{D}_{AU} \cup X^u$,
    $\mathcal{D}_U \leftarrow \mathcal{D}_U / X^{active}$ *# Update datasets*
    Update the target model T via minimizing the cross-entropy loss on D_L
  end

Total Loss

$$\mathcal{L}_{total} = \begin{cases} \mathcal{L}_{ce} + \mathcal{L}_{bce} & \text{if } \mathcal{D}_{AU} = \emptyset \\ \mathcal{L}_{ce} + \mathcal{L}_{bce} + \mathcal{L}_{em} + \lambda \mathcal{L}_t & \text{if } \mathcal{D}_{AU} \neq \emptyset \end{cases}$$

03. Methods

5. Flow

Algorithm 1: Our Proposed Algorithm for Open-set AL

Input:
  Labeled data $\mathcal{D}_L$, unlabeled data $\mathcal{D}_U$, number of AL cycles R, known categories K, per-cycle budget b, models F, C, T, and $\{G_i\}_{i=1}^K$
Process:
  $\mathcal{D}_{AU} \leftarrow \emptyset$ *# Initial active unknowns*
  Update models F, C, and $\{G_i\}_{i=1}^K$ by minimizing $\mathcal{L}_{total}$ in Eq. 9
  for $c = 0, 1, ..., R-1$ **do**
    $\forall x \in \mathcal{D}_U, \mathcal{S}_d(x) \leftarrow 0$ *# Initialization*
    if $\mathcal{D}_{AU} \neq \emptyset$ **do**
      Cluster the features of $\mathcal{D}_{AU}$ into K clusters
      For cluster i, compute the center $\mathbf{c}_i$ using Eq. 5
      $\forall x \in \mathcal{D}_U$, compute $\mathcal{S}_d(x)$ via Eq. 6
    end if
    $\forall x \in \mathcal{D}_U$, compute $\mathcal{S}_c(x)$ via Eq. 2
    Cluster the features of $\mathcal{D}_U$ into K clusters, $\{C_1, C_2, ..., C_K\}$ *# Diversity*
    for $j = 1, 2, ..., K$ **do**
      $\mathcal{S}_j \leftarrow \{\mathcal{S}_c(x) - \mathcal{S}_d(x) | \forall x \in C_j\}$ *# Uncertainty*
      $X_j \leftarrow$ select the $\frac{b}{K}$ samples with the lowest values from $\mathcal{S}_j$, and annotate them
    end
    # All queries for the current cycle:
    $X^{active} \leftarrow X_1 \cup X_2 \cup ... \cup X_K$
    # Knowns and active unknowns:
    Obtain X^k and X^u
    $\mathcal{D}_L \leftarrow \mathcal{D}_L \cup X^k, \mathcal{D}_{AU} \leftarrow \mathcal{D}_{AU} \cup X^u$,
    $\mathcal{D}_U \leftarrow \mathcal{D}_U / X^{active}$ *# Update datasets*
    Update the target model T via minimizing the cross-entropy loss on D_L
  end

Active Unknown 존재 시
- FINCH를 이용한 K Clustering 및 S_d 계산

Diversity를 반영하기 위해
- FINCH를 이용한 K Clustering

두개의 다른 Clustering
1). Active Unknown Clustering (S_d)
2). Known Class Clustering (Diversity)

03. Methods

5. Flow

Algorithm 1: Our Proposed Algorithm for Open-set AL

Input:
Labeled data $\mathcal{D}_L$, unlabeled data $\mathcal{D}_U$, number of AL cycles R, known categories K, per-cycle budget b, models F, C, T, and $\{G_i\}_{i=1}^{K}$
Process:
$\mathcal{D}_{AU} \leftarrow \emptyset$ *# Initial active unknowns*
Update models F, C, and $\{G_i\}_{i=1}^{K}$ by minimizing $\mathcal{L}_{total}$ in Eq. 9
for $c = 0, 1, ..., R-1$ **do**
  $\forall x \in D_U, S_d(x) \leftarrow 0$ *# Initialization*
  if $\mathcal{D}_{AU} \neq \emptyset$ **do**
    Cluster the features of $\mathcal{D}_{AU}$ into K clusters
    For cluster i, compute the center $\mathbf{c}_i$ using Eq. 5
    $\forall x \in D_U$, compute $S_d(x)$ via Eq. 6
  end if
  $\forall x \in D_U$, compute $S_c(x)$ via Eq. 2
  Cluster the features of $\mathcal{D}_U$ into K clusters, $\{C_1, C_2, ..., C_K\}$ *# Diversity*
  for $j = 1, 2, ..., K$ **do**
    $\mathcal{S}_j \leftarrow \{S_c(x) - S_d(x) | \forall x \in C_j\}$ *# Uncertainty*
    $X_j \leftarrow$ select the $\frac{b}{K}$ samples with the lowest values from $\mathcal{S}_j$, and annotate them
  end
  # All queries for the current cycle:
  $X^{active} \leftarrow X_1 \cup X_2 \cup ... \cup X_K$
  # Knowns and active unknowns:
  Obtain X^k and X^u
  $\mathcal{D}_L \leftarrow \mathcal{D}_L \cup X^k$, $\mathcal{D}_{AU} \leftarrow \mathcal{D}_{AU} \cup X^u$,
  $\mathcal{D}_U \leftarrow \mathcal{D}_U / X^{active}$ *# Update datasets*
  Update the target model T via minimizing the cross-entropy loss on D_L
end

1). S_c 계산 및 최종 Score 계산
2). Score가 낮은순으로 $\frac{b}{K}$ 개 씩 각 Known Class 별 Sampling
3). Labeler에게 전달

03. Methods

5. Flow

Algorithm 1: Our Proposed Algorithm for Open-set AL

Input:
  Labeled data $\mathcal{D}_L$, unlabeled data $\mathcal{D}_U$, number of AL cycles R, known categories K, per-cycle budget b, models F, C, T, and $\{G_i\}_{i=1}^{K}$
Process:
  $\mathcal{D}_{AU} \leftarrow \emptyset$ *# Initial active unknowns*
  Update models F, C, and $\{G_i\}_{i=1}^{K}$ by minimizing $\mathcal{L}_{total}$ in Eq. 9
  for $c = 0, 1, ..., R-1$ **do**
    $\forall x \in D_U, S_d(x) \leftarrow 0$ *# Initialization*
    if $\mathcal{D}_{AU} \neq \emptyset$ **do**
      Cluster the features of $\mathcal{D}_{AU}$ into K clusters
      For cluster i, compute the center $\mathbf{c}_i$ using Eq. 5
      $\forall x \in D_U$, compute $S_d(x)$ via Eq. 6
    end if
    $\forall x \in D_U$, compute $S_c(x)$ via Eq. 2
    Cluster the features of $\mathcal{D}_U$ into K clusters, $\{C_1, C_2, ..., C_K\}$ *# Diversity*
    for $j = 1, 2, ..., K$ **do**
      $\mathcal{S}_j \leftarrow \{S_c(x) - S_d(x) | \forall x \in C_j\}$ *# Uncertainty*
      $X_j \leftarrow$ select the $\frac{b}{K}$ samples with the lowest values from $\mathcal{S}_j$, and annotate them
    end
    # All queries for the current cycle:
    $X^{active} \leftarrow X_1 \cup X_2 \cup ... \cup X_K$
    # Knowns and active unknowns:
    Obtain X^k and X^u
    $\mathcal{D}_L \leftarrow \mathcal{D}_L \cup X^k, \mathcal{D}_{AU} \leftarrow \mathcal{D}_{AU} \cup X^u$,
    $\mathcal{D}_U \leftarrow \mathcal{D}_U / X^{active}$ *# Update datasets*
    Update the target model T via minimizing the cross-entropy loss on D_L
  end

Target Model Update

04

Experiments

04. Experiments

1. Main Results

- **CIFAR-10, CIFAR-100, TinyImageNet**
  - 초기 레이블링 데이터 : 1%, 8%, 8% 랜덤 샘플링
  - 20, 30, 40%를 Known Class로 설정, 나머지 Unknown Class
  - 매 Cycle마다 1,500장 Sampling 및 Annotation 요청

- **Implement Details**
  - Backbone : Resnet18 (No Pretrained Model)
  - AL Cycle : 300Epoch

- **Comparison Methods**
  - Standard AL
    1. Random Sampling
    2. Entropy
    3. Certainty
    4. BALD (Bayesian Active Learning by Disagreement)
    5. Coreset
    6. BADGE

  - Open-set AL
    1. MQNet
    2. LfOSA
    3. **EOAL(Proposal)**

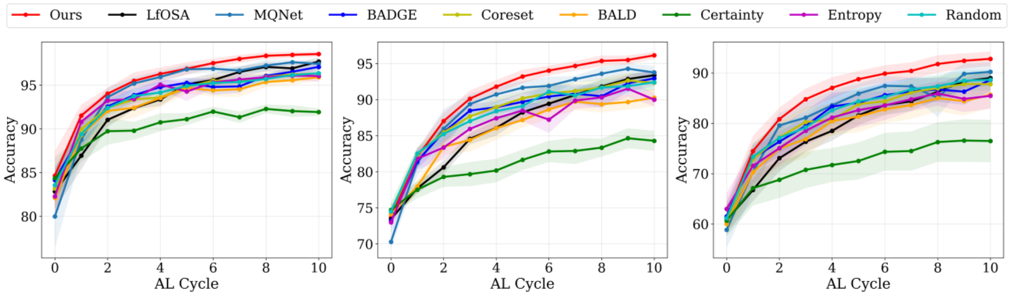


Figure 3: Classification accuracy comparison on CIFAR-10 (mismatch ratios from left to right: 20%, 30%, and 40%).

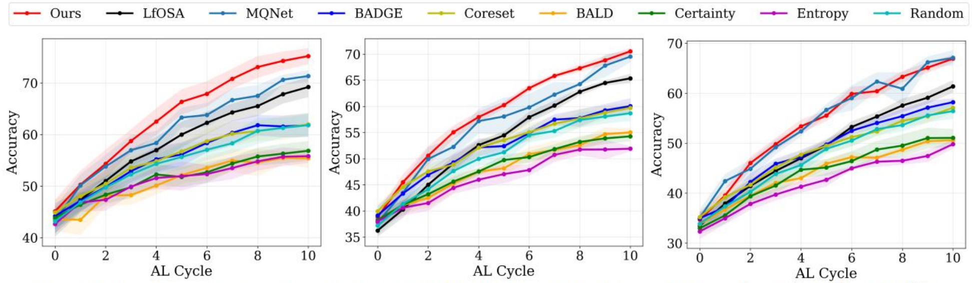


Figure 4: Classification accuracy comparison on CIFAR-100 (mismatch ratios from left to right: 20%, 30%, and 40%).

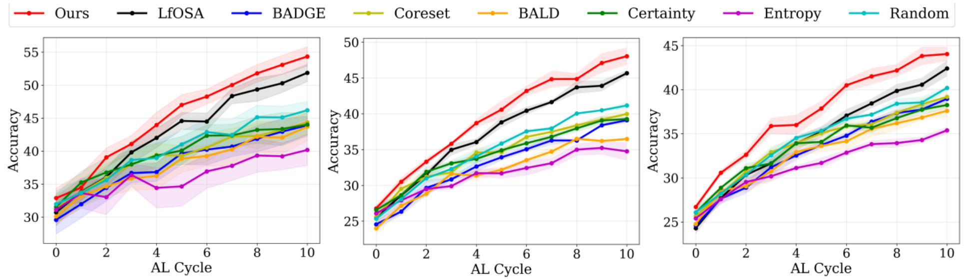


Figure 5: Classification accuracy comparison on TinyImageNet (mismatch ratios from left to right: 20%, 30%, and 40%).

04. Experiments

1. Main Results

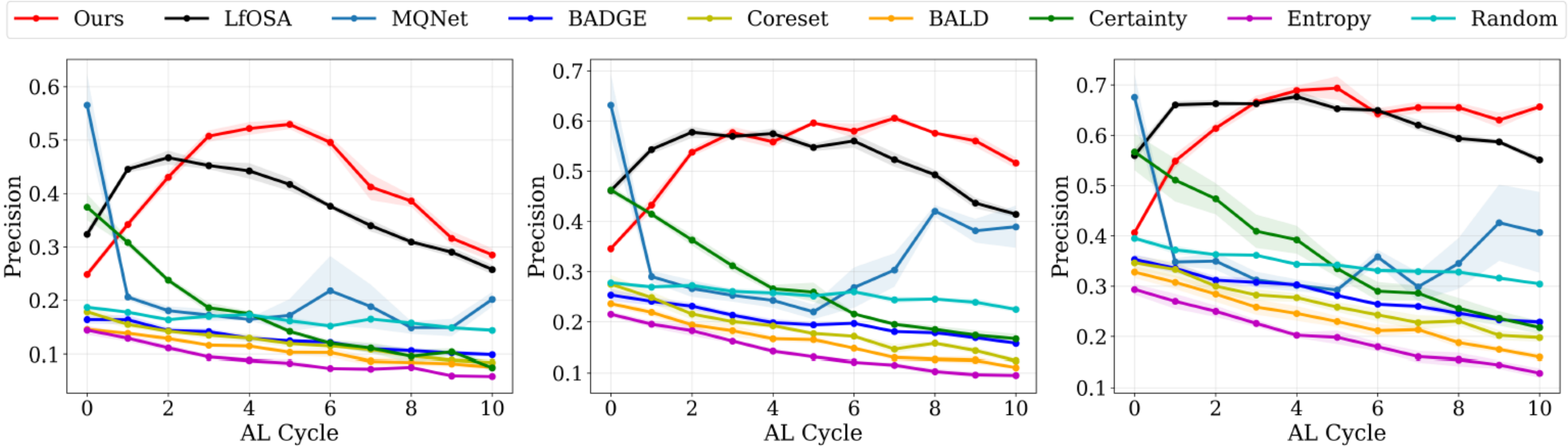
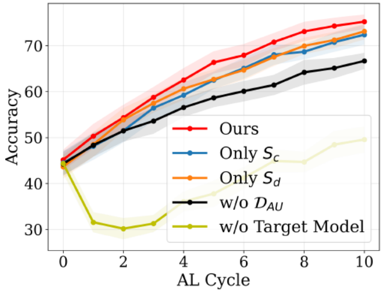


Figure 6: Precision results on CIFAR-100 (mismatch ratios from left to right: 20%, 30%, and 40%).

- **Known Class Sampling Precision**
  - LfOSA : Known Class Sample을 선택하는데 집중한 방법 → Known class Sampling에 집중한 방법
  - MQNet : Unknown Class의 비율이 높을 수 록 Known Class Selection 성능 저하
  - EOAL(Ours) : Known Class Sampling과 Diversity 둘다 확보

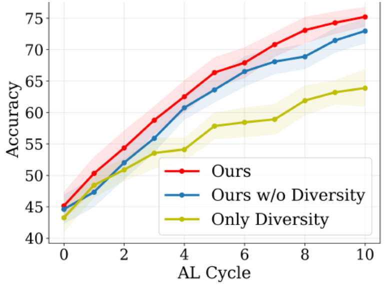
04. Experiments

2. Ablation Study

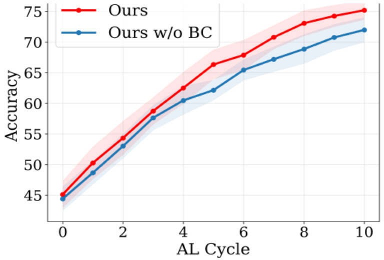


- **Effectiveness of Each Components**
  - Only S_c : S_c만으로 학습 시 성능 감소 → Unknown Class에 대한 고려 추가 필요
  - Only S_d : S_d만으로 학습 시 성능 감소 → Known Class에 대한 정보 부재
  - w/o $\mathcal{D}_{AU}$: Unknown sample을 사용하여 학습 X
  - w/o Target Model : Feature Extractor와 C로 평가

$$\mathcal{L}_{total} = \begin{cases} \boxed{\mathcal{L}_{ce} + \mathcal{L}_{bce}} & \text{if } \mathcal{D}_{AU} = \emptyset \\ \mathcal{L}_{ce} + \mathcal{L}_{bce} + \mathcal{L}_{em} + \lambda \mathcal{L}_t & \text{if } \mathcal{D}_{AU} \neq \emptyset \end{cases}$$



- **Diversity**
  - w/o Diversity(Clustering) : Cluster 별 Sampling 이 아닌 전체 중 Score 오름차순으로 선택
  - Only Diversity : Cluster 마다 Score 오름차순 선택이 아닌 Random Sampling



- **Binary Classifier**
  - w/o Binary Classifier : Classifier C를 통해 S_c 계산

05

Conclusion

05. Conclusion

- **Diversity까지 고려하여 학습에 필요한 데이터셋을 효과적으로 Sampling**

- **Method 선택 기준, 결과 해석 등의 설명이 다소 부족**

- **Query Strategy에서, Scoring 기준에 따라 Unknown Class Sampling도 가능 ?**

  - 실 세계에서는 Unknown Sampling 또한 추출하여 Known Sample로 편입시킬 필요 O
  - 따라서 Lifelong Learning 개념에서 Class Incremental 관점에서 본 논문이 기여할 수 있는 방향이 있지 않을까 ?

Q & A

Thank you for your attention